Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/22/2005
PI Name:	Dinges, David F. Ph.D.		
Project Title:	Countermeasures to Neurobehavioral Deficits from Cumulative Partial Sleep Deprivation During Space Flight		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Human Performance Factors, Sleep, and Chronobiology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	10/01/2000	**End Date:**	06/30/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	68
No. of Bachelor's Candidates:	38	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	VanDongen, Hans (University of Pennsylvania Health System) Rogers, Naomi (University of Pennsylvania)		
Grant/Contract No.:	NCC 9-58-HPF00204		
Performance Goal No.:			
Performance Goal Text:			
	This project is concerned with identifying methods to prevent neurobehavioral and physical deterioration due to inadequate sleep and sleep placed at different times across the 24-hour day in astronauts during long-duration manned space flight. The performance capability of astronauts during extended-duration space flight depends heavily on achieving recovery through adequate sleep. Even with appropriate circadian alignment, sleep loss can erode fundamental elements of human performance capability including vigilance, cognitive speed and accuracy, working memory, reaction time, and physiological alertness. When attempting to sleep and perform at an adverse circadian phase, the magnitude and time course of sleep loss and consequent deficits in neurobehavioral functioning are significantly affected. Adequate sleep is essential during manned space flight not only to ensure high levels of safe and effective human performance, but also as a basic regulatory biology critical to healthy human functioning.		

Task Description:	There is extensive objective evidence that astronaut sleep is restricted in space flight to averages between 4 hr and 6.5 hr/day. Chronic sleep restriction during manned space flight can occur in response to endogenous disturbances of sleep (motion sickness, stress, circadian rhythms), environmental disruptions of sleep (noise, temperature, light), and curtailment of sleep due to the work demands and other activities that accompany extended space flight operations. The mechanism through which this risk emerges is the development of cumulative homeostatic pressure for sleep across consecutive days of inadequate sleep. Research has shown that the physiological sleepiness and performance deficits can progressively worsen (i.e., accumulate) over consecutive days of sleep restriction, and that sleep limited to levels commonly experienced by astronauts (i.e., 4 - 6hr per night) for as little as 1 week, can result in increased lapses of attention, degradation of response times, deficits in complex problem solving, reduced learning, mood disturbance, disruption of essential neuroendocrine, metabolic, and neuroimmune responses, and in some vulnerable persons, the emergence of uncontrolled sleep attacks. The prevention of cumulative performance deficits and neuroendocrine disruption from sleep restriction during extended duration space flight involves finding the most effective ways to obtain sleep in order to maintain the high-level cognitive and physical performance functions required for manned space flight. There is currently a critical deficiency in knowledge of the effects of how variations in sleep duration and timing relate to the most efficient return of performance per unit time invested in sleep during long-duration missions, and how the nature of sleep physiology (i.e., sleep stages, sleep electroencephalographic [EEG] power spectral analyses) changes as a function of sleep restriction, the timing of sleep, and performance degradation. The primary aim of this project is to meet these critical deficiencies through utilization of a response surface experimental paradigm. Through testing in a dose-response manner, varying combinations of sleep duration and timing, this project will help establish how to most effectively limit the cumulative adverse effects of chronic sleep restriction in space operations on human performance and physiology. Although there is evidence that the less sleep obtained, the greater the waking deficits, experiments have found that for acute periods supplementing a reduced anchor sleep period with a nap has the potential to enhance performance, due to the exponential recovery of neurobehavioral functions relative to sleep duration. During the past 5 years we have been using a response surface experimental approach to systematically determine the chronic (10-day) effects of 18 sleep schedule conditions. There are two experiments in this project. The first experiment involved restricted nocturnal anchor sleep alone and in combination with varying durations of restricted daytime naps on performance, mood, sleep, circadian physiology and hormones. The resulting preliminary response surface maps (RSMs) derived from this dose-response experiment indicate that total sleep time per 24hr is a prime determinant of cumulative neurobehavioral deficits, and that combining a restricted nocturnal anchor sleep with a midday nap can attenuate cumulative deterioration in performance. In order to complete our understanding of how to optimize performance in the face of restricted sleep in space flight, in the second experiment we have reversed the circadian placement of these 18 anchor sleep + nap sleep conditions (i.e., daytime anchor sleep alone and in combination with varying durations of restricted nocturnal naps. To develop the response surface models, both experiments will require n=90 (total N=180) healthy men and women to undergo a 14-day ground-based laboratory protocol involving random assignment to one of 18 sleep-ration cells. The 18 sleep ration cells utilized in experiment 1 (nocturnal anchor sleep) will be repeated in experiment 2 (diurnal anchor sleep) for a total of 36 sleep ration cells. The sleep-ration assignments involve 4 anchor sleep durations (4.2, 5.2, 6.2, 8.2 hr) and 6 nap sleep durations (0.4, 0.8, 1.2, 1.6, 2.0, 2.4 hr) crossed to yield a total of 4 anchor-sleep-only conditions, and 14 anchor + nap sleep conditions, and spanning a dynamic range of cumulative sleep debts (i.e., from 0 to 40 hr in a 10-day period). Subjects undergo a wide range of quasi-continuous neurobehavioral performance tests and continuous physiological monitoring of waking EEG, sleep PSG, behavioral motility, and body temperature, while living in the laboratory for 14 consecutive days. The laboratory environment is designed to simulate the low light, tight quarters, and lack of social contact with the outside world that will characterize long-duration space flight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Sleep loss, in particular chronic sleep loss, is becoming increasingly more common in today's global society. Chronic partial sleep loss without adequate recovery sleep leads to what is referred to as "sleep debt". Weekly restriction of sleep to <6.5h per night is common in many segments of society including shift workers; long-haul truck drivers; police personnel; medical workers; transoceanic pilots; and astronauts. In addition, recent epidemiological studies reported that between 15-20% of Americans sleep 6.5h or less per night. Similar estimates have been reported in other populations in Europe, Asia and Australia. Sleep restriction is associated with increased risk of errors, traffic accidents, injuries, interpersonal conflicts, stimulant use (licit and illicit), and burnout. These behavioral problems are based in altered brain functions due to chronic sleep restriction. Recent laboratory-based, randomized controlled experiments provide extensive evidence that chronic restriction of sleep to 4h, 5h or 6h per day, for periods from 5 to 14 days, results in cumulative neurobehavioral impairments, irrespective of the circadian phase at which the sleep is obtained. Cognitive performance deficits accumulate across days of partial sleep loss to levels equivalent to those found after 1 to 3 nights of total sleep loss. Sleep dose-dependent cumulative effects were also observed in a recent 7-day, laboratory study of truck drivers. In addition to the behavioral effects, chronic sleep loss also poses significant health risks. Epidemiological studies have found that short sleep (<6.5h per night) is associated with increased risk of coronary heart disease, and with an overall increased risk of mortality. Laboratory studies of healthy adults subjected to chronic partial sleep loss have found adverse effects on endocrine functions, metabolic responses, and immune inflammatory responses. One potential countermeasure to minimize the risk of these neurobehavioral and physiological alterations is to supplement shortened sleep durations with additional, short sleep periods, or naps. This may be especially beneficial when a consolidated nocturnal sleep period is not possible, and individuals are required to be awake across the night and sleep during the day, for example shift workers, and during adjustment to new time zones following transmeridian travel. It is important to determine what duration of nap will provide the most benefit for alertness and cognitive functions, while still avoiding the detrimental effects of sleep inertia following termination of the sleep period. Information obtained from the current research project will provide important information on the effects of chronically restricted sleep, placed at an adverse circadian phase (i.e. diurnally), with and without nocturnal naps, on neurobehavioral and physiological functioning, including sleep physiology, endocrine measures and thermoregulation. In addition to providing information on the effectiveness and effects of different split sleep-wake schedules for use in spaceflight, this information will also be applicable for a large number of Earth based industries and individuals who are chronically exposed to sleep restriction. Ultimately this will help improve health and safety of individuals on Earth.

Task Progress:	During the final year of the project we concluded data collection on experiment 2, so that all 18 diurnal anchor sleep plus nocturnal nap conditions were completed. A total of N=79 subjects completed the 15 day (14 night) in-laboratory protocol, for a total of 1,185 24 hour in-laboratory study days. Preliminary analyses have been completed, indicating that the findings from our second study examining the effects of restricted diurnal anchor sleeps with nocturnal naps support the findings from our first experiment, with restricted nocturnal anchor sleep periods and diurnal nap periods. Subjects were able to achieve physiological sleep on split-sleep schedules, and naps helped attenuate some cumulative impairments, but TST per 24h (by PSG) remained the prime determinate of behavioral capability over time. Moreover, naps taken at the nocturnal circadian phase, resulted in severe sleep inertia upon awakening, which had to dissipate before nap benefits on performance were seen. Further analyses of the cognitive performance tasks and subjective scales is continuing, including the development of RSMs for these variables. These RSMs will then be compared to and combined with those generated from the first experiment to examine the effects of chronic sleep restriction with sleep placed at different circadian phases. Analysis of the polysomnographic data is continuing, and when complete will be used to generate RSMs for various sleep physiology variables, including total sleep time, sleep efficiency and sleep stages. In order to determine the rate and degree of circadian adaptation to the abrupt reversal of the sleep-wake schedule core body temperature will be analyzed, in addition to melatonin profiles in those subjects where blood sampling was done. The effects of chronic sleep restriction and circadian disruption on cortisol secretion will be analyzed in those subjects with blood samples collected. Significant Media Coverage: Proceedings of the Fatigue and Performance Modeling Workshop, Aviation, Space Environmental Medicine, 75(3 Supplement) March 2004. National Geographic cover for special edition on Space Travel
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